

Functions and Function Notation (Homework)

① If $f(x) = 4 - x^2$, find i) $f(1)$, ii) $f(2)$, and iii) $f(-3)$.

② If $f(x) = \sqrt{3x-1}$, find i) $f(10)$, ii) $f(1)$, and iii) $f(3)$.

③ If $f(x) = 2^{x+4}$, find i) $f(0)$, ii) $f(-1)$, and iii) $f(-4)$.

④ If $f(x) = |x+5| - 2$, find i) $f(6)$, ii) $f(0)$, and iii) $f(5)$.

⑤ If $f(x) = \frac{3}{x^4}$, find i) $f(2)$, ii) $f(1)$, and iii) $f(0)$.

⑥ If $f(x) = 2 + x$, fill in the table below.

x	f(x)
0	
-6	
π	
$\frac{3}{4}$	

⑦ If $f(x) = 10 + \sqrt{x}$, fill in the table below.

x	9	36	18	-4
f(x)				

→ Hint: Functions do not have imaginary outputs.

⑧ If $f(x) = \frac{2}{x-1} + x^3 + 6$, fill in the table below.

x	f(x)
0	
5	

⑨ If $f(x) = 2x^2 - x + 7$, fill in the table below.

x	-1	3	2	-10
f(x)				

⑩ If $f(x) = \frac{1}{4x}$, find i) $f(b)$, ii) $f(0.2)$, and iii) $f(\frac{1}{4})$.

⑪ If $f(x) = -10x^2 + 5x$, find i) $f(a+1)$, ii) $f(\pi)$, and iii) $f(c)$.

⑫ If $f(x) = x - 100 - \sqrt{3x+2}$, find i) $f(2.5)$, ii) $f(3^{b+1})$, and iii) $f(0)$.

In the tables below, x represents the inputs and y represents the outputs. Circle the tables that could not represent functions.

⑬

x	y
3	9
2	8
1	7
0	6
-1	5
-2	4

x	y
0	-1
1	10
7	5
2	-5
3	4
1	9

x	y
11	14
0	17
8	19
3	7
20	21
6	13

x	y
100	25
200	50
300	75
400	100
500	125
600	150

⑭

x	-6	4	3	-8	5	6
y	8	4	5	6	7	9

x	9	10	4	3	-1	2
y	3	5	6	2	0	1

x	14	12	20	14	30	32
y	13	20	11	15	16	0

x	13	15	10	11	17	17
y	0	1	2	3	4	5

(15)

X	$\frac{1}{2}$	0	3	$4\frac{1}{3}$	$2\frac{1}{5}$	5
Y	7	20	4	13	8	6

X	-1	1	-2	-3	7	4
Y	5	5	0	8	10	12

X	Y
14	6
15	7
16	6
17	7
18	6
19	7

X	Y
13	1
20	3
19	5
18	7
21	8
13	9

(16)

x	y
10	7
3	4
8	19
10	7
10	6
7	8

x	y
13	15
10	15
18	15
20	15
25	15
26	15

x	y
8	4
2	6
7	9
6	11
8	13
9	15

x	y
16	1
17	1
17	2
19	1
20	1
20	-1

(17)

X	3	8	8	6	-2	-1
Y	10	2	2	0	4	5

X	7	4	7	2	7	8
Y	3	5	3	5	3	8

X	10	3	8	10	11	11
Y	4	4	9	4	9	10

X	-4	-2	4	3	-2	7
Y	7	8	9	9	8	9

(18)

X	5	3	0	-1	-1	-1
Y	7	6	2	4	5	4

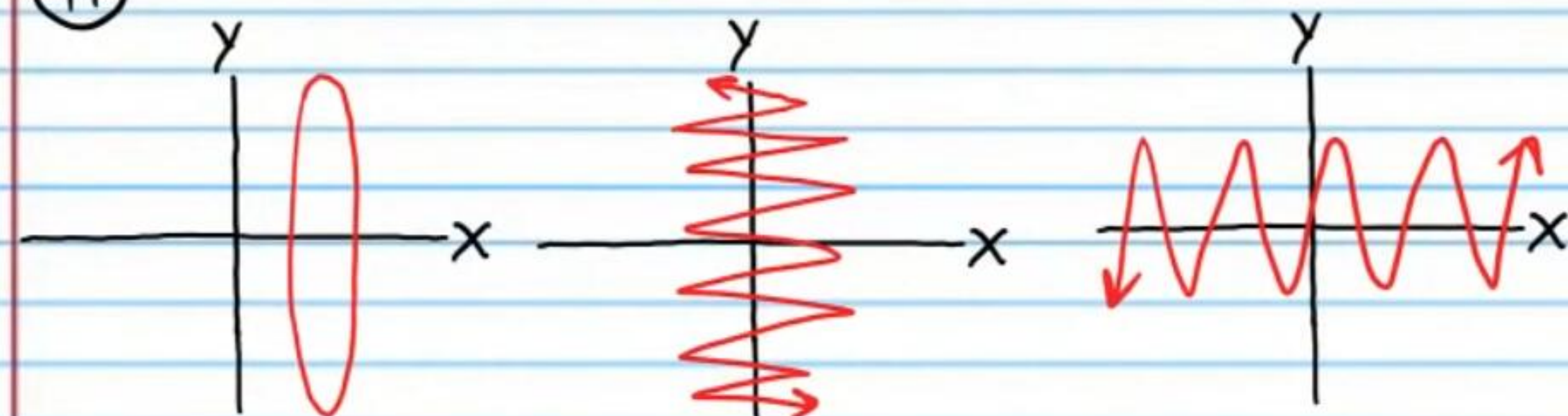
X	7	5	9	7	11	11
Y	7	7	7	4	7	7

x	y
0	0
1	13
2	15
3	15
1	13
0	100

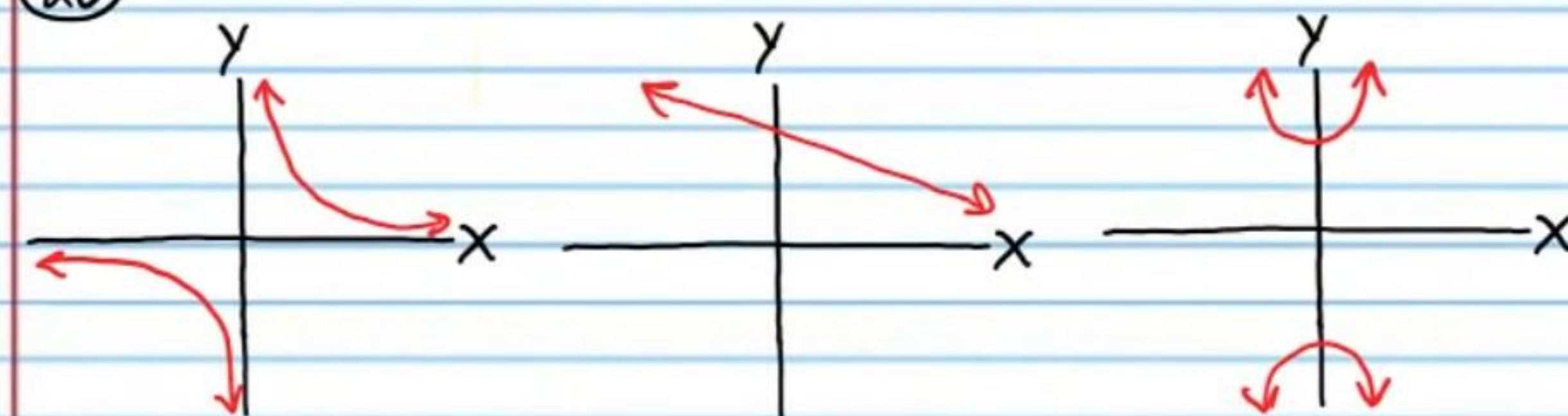
x	y
16	-8
10	-8
8	-3
0	-1
8	1
0	3

Circle the relations below that cannot be functions.

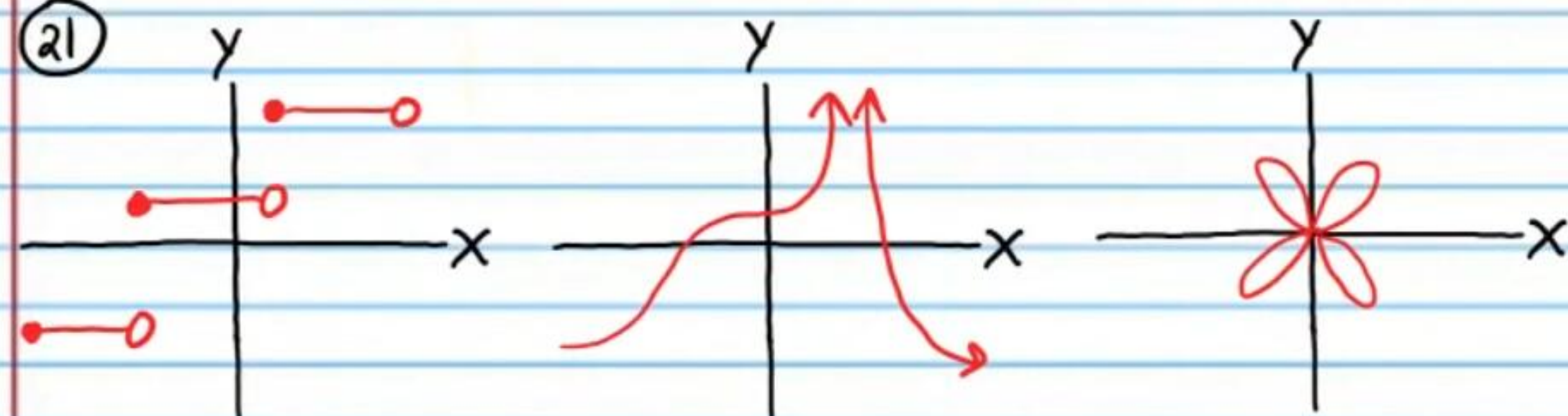
(19)



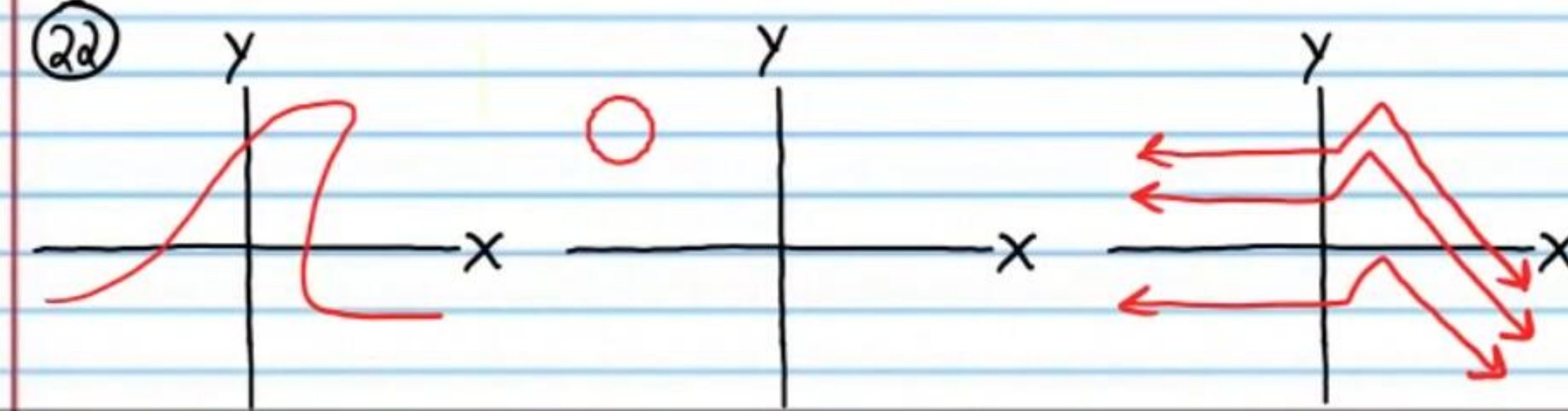
(20)



(21)



(22)



Answers

① i) $\boxed{3}$ ii) $\boxed{0}$ iii) $\boxed{-5}$

② i) $\boxed{\sqrt{29}}$ ii) $\boxed{\sqrt{2}}$ iii) $\sqrt{8} = \boxed{2\sqrt{2}}$

③ i) $\boxed{16}$ ii) $\boxed{8}$ iii) $\boxed{1}$

④ i) $\boxed{9}$ ii) $\boxed{3}$ iii) $\boxed{8}$

⑤ i) $\boxed{\frac{3}{16}}$ ii) $\boxed{3}$ iii) $\boxed{\text{undefined}}$

⑥

x	f(x)
0	2
-6	-4
π	$2+\pi$
$\frac{3}{4}$	$2\frac{3}{4}$

⑦

x	9	36	18	-4
f(x)	13	16	$10+3\sqrt{2}$	und

⑧

x	f(x)
0	4
5	$13\frac{1}{2}$

⑨

x	-1	3	2	-10
f(x)	10	22	13	217

⑩ i) $\boxed{\frac{1}{4b}}$ ii) $\boxed{1.25 \text{ or } \frac{5}{4}}$ iii) $\boxed{1}$

⑪ i) $\boxed{-10(a+1)^2 + 5(a+1)}$ ii) $\boxed{-10\pi^2 + 5\pi}$ iii) $\boxed{-10c^2 + 5c}$

⑫ i) $\boxed{-97.5 - \sqrt{9.5}}$ ii) $\boxed{3^{b+1} - 100 - \sqrt{3(3^{b+1}) + 2}}$ iii) $\boxed{-100 - \sqrt{2}}$

⑬ $\square \square \square \square$

⑭ $\square \square$

⑮ $\square \square$

⑯ $\square \square \square \square$

⑰ $\square \square$

⑱ $\square \square \square$

⑲ $++$

⑳ $++$

㉑ $++$

㉒ $+++$